Marmara University – Faculty of Engineering – Department of Computer Engineering

Fall 2020 – CSE3055 Database Systems
Quiz #3

(31.12.2020 | Duration: 45 min)

Consider the *University* database below.

Student

SID	SName	Major	Year	Age
100	Jones	History	GR	21
150	Parks	Accounting	SO	19
200	Baker	Math	GR	50
250	Glass	History	SR	50
300	Baker	Accounting	SR	41
350	Russell	Math	JR	20
400	Rye	Accounting	GR	18
450	Jones	History	FR	24

Course

CCode	CTime	Room	NbrSt	AvgTotal
BA200	MWF9	SC110	3	
BD445	MWF3	SC213	3	
BF410	MWF8	SC213	1	
CS150	MWF3	EA304	2	
CS250	MWF12	EB210	2	

Enrollment

SID	CCode	Midterm	Final	Total
100	BD445	70	77	
150	BA200	65	89	
200	BD445	71	93	
200	CS250	93	98	
300	CS150	88	99	
400	BA200	100	78	
400	BD445	54	68	
400	BF410	47	69	
400	CS150	33	55	
400	CS250	86	87	
450	BA200	79	81	

Notes:

- *Course.NbrSt*: number of students taking that course.
- *Course.AvgTotal*: average of *Totals* of student-course enrollments from table *Enrollment* for a related course.
- *Enrollment.Total*: summation of 40% of *Midterm* and 60% of *Final*.

1) [60 pts] Implement a stored procedure *sp_UpdateAvgTotal* (*Course.CCode*) such that

- *exec sp_UpdateAvgTotal* (*Course.CCode*) updates the field *AvgTotal* of table *Course*.
  - The field *AvgTotal* is the average of *Totals* of student-course enrollments from table *Enrollment* for a related course. In order to calculate field *AvgTotal* of table *Course*, field *Total* of table *Enrollment* is first to be computed by adding 30% of *Midterm* and 70% of *Final*.
- Create a table for logging errors before creating *sp_UpdateAvgTotal*. Name the table as *Log*, table will have three fields: *LogID, CCode, ErrCode*.
  - The primary key will be *LogID*.
  - The field *LogID* will be an identity column, *CCode* will be corresponding *CCode* that is to be entered, and *ErrCode* will be "*I*" for "calling invalid course" (*i.e.* not available in table *Course*), "*NC*" for "no enrolled course" (*i.e.* not available in table *Enrollment*).
  - You have to enter appropriate records to the table *Log*. After these errors, the transaction will be rolled back.
  - Data types for the fields are as follows: *LogID* int; *CCode* varchar(8); *ErrCode* varchar(2).

2) [40 pts] Write a trigger that implements the following:

- When a record of student (*SID*) and course (*CCode*) is inserted, deleted or updated from the table *Enrollment*, update the field *NbrSt* field of table *Course*.
- The field *NbrSt* is the number of students that is enrolled to the related course.
- Hint: you can assume there is only one insertion, deletion or update at a time.

IMPORTANT NOTES:

1) Textbooks, slides and notes are closed.

2) Internet usage is not allowed, however computers and cell phones may be used for only viewing this exam.

3) This exam consists of 2 questions and 2 pages.

4) Please write your answers clearly and neatly to the appropriate place on blank A4 pages.

5) Show all your work.

6) Write your student ID, first name and last name on top of each solution page.

7) Write the following sentences on the first page of your document: "I hereby swear that the work done on this quiz is totally my own; and on my honor, I have neither given nor received any unauthorized and/or inappropriate assistance for this quiz. I understand that by the school code, violation of these principles will lead to a zero grade and is subject to harsh discipline issues." Write your full name and sign it.

8) Solve all the questions in your handwriting on blank A4 sheets.

9) Your solution pages should be in the order given in this document.

10) Scan all the solution pages to a single PDF file. Your scans should be clear, readable, small in size, cropped and fixed, portrait oriented, upright and pages ordered.

11) Name your scanned file as "StudentID.pdf".

12) Upload your PDF file to http://ues.marmara.edu.tr before deadline.

13) In case of any form of copying and cheating on solutions, all parts will get ZERO points. You should submit your own work. In case of any forms of cheating or copying, both giver and receiver are equally culpable and suffer equal penalties. All types of plagiarism will result in zero points from the quiz.

14) Do not send quiz submissions through e-mail or any other media. E-mail attachments or such will not be accepted as valid submissions.

15) You are responsible for making sure you are turning in the right file, and that it is not corrupted in anyway. We will not allow resubmissions if you turn in the wrong file, even if you can prove that you have not modified the file after the deadline.

16) No groups are allowed.

17) Grade evaluation may be done on selected parts of the quiz, so try to complete all parts of your quiz successfully.

18) No late submissions will be accepted.